MIL-D-51082E
12 October 1984
SUPERSEDING
MIL-D-0051082D(MU)
15 April 1969
MIL-D-51082C
29 December 1966

MILITARY SPECIFICATION

DETECTOR TUBES, CHEMICAL AGENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers detection tube assemblies consisting of clips of tubes for sampling and detecting chemical agents.

1.2 Classification. The tubes shall be of the following types:

Type I	-	Mustard and G-agent (H and G) detector
Type II	-	Hydrogen cyanide (AC) detector
Type III	-	Phosgene (CG) detector
Type IV	-	Arsenicals (L and ED) detector
Type V	-	Arsenical (L) detector
Type VI	-	CN-BBC detector
Type VII	-	Sampling tube

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

FSC 6665

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SPECIFICATIONS

MILITARY

MIL-S-14374 - Silica Gel, Iron Free
MIL-P-50014 - Packet, Sodium Pyrophosphate Peroxide
MIL-T-51023 - Tablets, Compound 34

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

5-77-387 - Clips, Detector Tube
5-77-446 - Tubes, Detector
5-77-784 - Dispenser, Filled
5-77-820 - Clips, Detector Tube
5-77-2509 - Detector Tube Assembly
IL 136-41-780 - Dilution Apparatus, Q5, Single and Double Dilution

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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ASTM STANDARDS

D 1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

MUNSELL BOOK OF COLOR

(The Munsell Book of Color may be obtained from the Munsell Color Company, Inc., 2441 N. Calvert Street, Baltimore, MD 21218.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-77-446 and on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the detector tubes shall conform to the specifications and drawings listed on Drawing 5-77-446.

3.2 Manufacture and assembly. The detector tubes shall be manufactured in accordance with Drawing 5-77-446 and assembled in accordance with Drawing 5-77-387, 5-77-784, 5-77-820, or 5-77-2509, as applicable (see 6.2).

3.2.1 Gels. Silica gel conforming to MIL-S-14374 shall be used and impregnated with the applicable materials indicated in table I (see 6.4). Impregnated gels shall be no older than 120 days when sealed into detector tubes (all type tubes except type VII). Sealed tubes shall be no older than 120 days when the 3-year expiration date is stamped on the kit.

3.2.2 Charcoal. Forty to sixty mesh activated charcoal shall be used in type II tubes (see 6.4).

3.3 Leakage. The detector tubes shall not leak when tested in accordance with 4.4.4.1.

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TABLE I. Silica gel impregnation materials

Tube type	Impregnation
I	DB3 Mercuric cyanide
II	Cupric sulfate p,p'-Tetramethyldiaminodiphenylmethane Salicylic acid Dye
III	p-Dimethylaminobenzaldehyde N-Phenyl-1-naphthylamine Thiourea
IV	Ammonium molybdate Zinc sulfate
V	Cupric nitrate Potassium iodide
VI	m-Dinitrobenzene
VII	Silica gel (not impregnated)

3.4 Airflow resistance. All types of detector tubes shall have an airflow resistance of 40 to 70 millimeters (mm) of water. As an exception, type II detector tubes for the M18A2 detector kit and the M272 water testing kit shall have an airflow resistance of 40 to 95 mm of water. Test for airflow resistance shall be conducted as specified in 4.4.4.2.

3.5 Volatile matter. The volatile content of the gels and the charcoal in the detector tubes shall conform to table II when tested as specified in 4.4.4.3.

TABLE II. Volatile matter

Tube type	Percent by weight	
	Minimum	Maximum
I, IV, V	--	5.0
II Gel	--	15.0
Charcoal	14.0	--
III, VI	--	10.0
VII	--	5.0

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3.6 Stability. Before and after aging the detector tubes for 2 weeks at 65° ± 2°C, the gel layer shall comply with the designated colors in table III when tested in accordance with 4.4.4.4. The Munsell Book of Color shall be used in determining the color.

TABLE III. Munsell colors

Tube type	Hue	Value	Chroma
I, V, VI, VII	Neutral	9,9.5	0
II	5.0 R	9	0.5
	10 R	9	1
	5.0 YR	9	1
	10 YR	9	1,2
	2.5 Y	9	1,2
	5.0 Y	9	0.5,1,2
	7.5 Y	9	1,2
	5.0 GY	8,9	1
	5.0 G	9	0.5
III	5.0 Y	9	10,12,14
	5.0 Y	8	8,10,12
	7.5 Y	8,9	8,10
	7.5 Y	9	4
	7.5 Y	9	6
IV	7.5 Y	9	3,4,6,8
	10 Y	8	6,8
	2.5 GY	8	4,6
	2.5 GY	9	4

3.7 Sensitivity. Before and after aging for 2 weeks at 65° ± 2°C, the detector tubes shall comply with the requirements of table IV when tested as specified in 4.4.4.5.

3.8 Score breakage. When tested in accordance with 4.4.4.6, the tubes shall break along the score mark without splintering when subjected to a maximum load of 950 grams (g). The tubes shall not break when subjected to a load of 400 g. When broken by hand, at least 60 percent of the tube shall break cleanly around the score mark and the remaining portion shall not have protrusions exceeding 1/32 inch.

3.9 Preproduction. Prior to the start of regular production, a preproduction lot of detector tubes shall be produced for examination and test in accordance with 4.3.

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TABLE IV. Sensitivity

Tube type	Requirement
Blanks for I, IV, V, VI	2.5Y to 7.5Y 9/3 (max)
I (H)	blue or purple-blue ring in 15 sec
I, VII (G)	yellow ring in 30 sec
II	blue ring immediately
III	green color immediately
IV	blue or blue-green ring in 15 sec
V	red-brown ring in 15 sec
VI	pink or magenta ring in 15 sec

3.10 Workmanship. The detector tubes shall not be cracked, chipped, or broken. They shall be free from foreign matter and shall be uniform in thickness with sealed ends which are smoothly rounded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of 300 of each type of detector tube shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production (see 4.4.1).

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive test. Sampling for each type of detector tube shall be conducted in accordance with MIL-STD-105. For leakage test, level S-4 shall be used.

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4.3.2.2 For destructive tests. Sampling for each type of detector tube shall be conducted in accordance with table V.

TABLE V. Sampling for destructive tests

Requirement	Sample size	Acceptance number	Test paragraph
Leakage	MIL-STD-105, level S-4	AQL 1.0 percent defective	4.4.4.1
Airflow resistance			4.4.4.2
Dwg 5-77-784	MIL-STD-105, level S-4	AQL 2.5 percent defective	
Dwg 5-77-820	MIL-STD-105, level S-4	AQL 2.5 percent defective	
Dwg 5-77-387	MIL-STD-414	AQL 0.1 percent variability unknown, standard deviation method, double specification limit	
Volatile matter	48 tubes (for lot size of 10,000 and under)	0 defects	4.4.4.3
	96 tubes (for lot size over 10,000)	1 defect	4.4.4.3
Stability and sensitivity	50 tubes*	3 defects	4.4.4.4, 4.4.4.5
Score breakage			
Machine break	MIL-STD-105, level S-4	AQL 2.5 percent defective	4.4.4.6(a)
Hand break	MIL-STD-105, level S-4	AQL 2.5 percent defective	4.4.4.6(b)

*Twenty-five tubes shall be tested for stability unaged and sensitivity unaged. Twenty-five tubes shall be tested for stability aged, and sensitivity aged.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive test. The sample detector tubes of each type shall be examined for all requirements of the applicable drawings, the classification of defects, and MIL-STD-105. Test shall be conducted as specified in 4.4.4.1 and table V.

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4.3.3.2 For destructive tests. Sample detector tubes of each type shall be tested as specified in 4.4.4.2 through 4.4.4.6 and table V.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the tubes of the same type assembled into the same size clip by one manufacturer from the same materials and under essentially the same manufacturing conditions. However, one lot of tubes shall contain no more than one lot of gel and shall have been filled within 5 consecutive working days. All tubes shall be sealed the same day they are filled.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105. For leakage test, level S-4 shall be used.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with table V.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. Sample detector tubes of each type shall be examined in accordance with the classification of defects and MIL-STD-105. Tests shall be conducted as specified in 4.4.4.1 and table V.

4.4.3.2 For destructive tests. Sample detector tubes of each type shall be tested as specified in 4.4.4.2 through 4.4.4.6 and table V.

4.4.3.3 Classification of defects.

(a) Tubes, detector, assembly and detail (Drawing 5-77-446).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Component missing or incorrectly assembled
102		Gel length incorrect
103		Length of charcoal incorrect (type II only)
104		Scored line missing, incomplete, or incorrectly located
105		Workmanship (3.10)
Minor:	AQL 2.5 percent defective	
201		Overall tube length incorrect
202		Color bands missing or incorrectly located
203		Paint sticky or tubes stuck together

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(b) Clips, detector tube (Drawing 5-77-387).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Number of tubes incorrect
102	Marking incorrect or missing
103	Workmanship (3.10)

(c) Dispenser, filled (Drawing 5-77-784).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Number of tubes incorrect
102	Marking incorrect or missing
103	Workmanship (3.10)

(d) Clips, detector tube (Drawing 5-77-820).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Number of tubes incorrect
102	Marking incorrect or missing
103	Workmanship (3.10)

(e) Packaging (see section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Container not as specified
102	Container damaged
103	Marking incorrect, missing or illegible
104	Repair part packaging and marking not as specified (when applicable)

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4.4.4 Tests. See 6.3 for testing precautions. Water in accordance with ASTM D 1193, type III, and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.4.4.1 Leakage. Place the tubes in a vacuum desiccator and apply a vacuum of 5 inches (minimum) of mercury. Close the stopcock and maintain the vacuum for 30 minutes. At the end of this period, draw in sufficient colored water (0.1 g 3,3'-dibromothymolsulfonphthalein and 8.0 milliliters (mL) N/50 sodium hydroxide per 2.5 liters) to cover the tubes. Release the vacuum. Wait 5 minutes and examine each tube for evidence of water penetration.

4.4.4.2 Airflow resistance. Determine the airflow resistance of each tube by measuring the pressure drop (in millimeters of water) across the inlet and outlet of the tube at a flow rate of 100 ± 5 mL of air per minute. The outlet of the tube shall be at atmospheric pressure.

4.4.4.3 Volatile matter. Remove the gel from 16 tubes. Composite and weigh to the nearest milligram (mg) in a tared weighing bottle. Dry in an oven at 150° ± 5°C for 3 hours ± 5 minutes. Determine the loss in weight. Perform the test on each group of 16 tubes.

4.4.4.4 Stability. Examine the gel layer of each tube and record its hue, value, and chroma. The Munsell color shall be read using a 7400°K light source in an enclosure having a neutral gray-colored interior. Continue as directed in 4.4.4.5.

4.4.4.5 Sensitivity. Determine the sensitivity using a flow rate of 1 liter per minute and an exposure time of 30 to 60 seconds (see 6.3).

(a) Apparatus and reagents.

Dilution Apparatus, Toxic Gas, Q5, (Drawing IL 136-41-780)
Sodium hydroxide solution - 8.5 ± 0.5 percent
Compound 34-pyrophosphate peroxide solution:
  2.5 mg compound 34 conforming to MIL-T-51023
  200 mg sodium pyrophosphate peroxide conforming to MIL-P-50014
  8 mL water

(b) Blank. Select eight tubes at random from the sample. Break off the ends of each tube and add 2 drops of sodium hydroxide solution to each tube. After 2 minutes, determine the Munsell color of the gel layer using a 7400°K light source in an enclosure having a neutral gray-colored interior.

(c) Type I. Use half of the tubes for each test.

(1) Sensitivity to mustard. Break off the ends of each tube and expose the gel to 0.5 microgram H. Remove the tube from the apparatus, wait 2 minutes, then add 2 drops of sodium hydroxide solution. Examine the gel layer. Record the color and reaction time.

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(2) Sensitivity to G-agent. Break off the ends of each tube and expose the gel to 1.0 microgram GB. Remove the tube from the apparatus and add 2 drops of compound 34-pyrophosphate peroxide solution. Examine the gel layer. Record the color and reaction time.

(d) Type II. Break off the ends of each tube and expose the gel to 8.0 micrograms AC. Remove the tube from the apparatus and examine the gel layer. Record the color and reaction time.

(e) Type III. Break off the ends of each tube and expose the gel to 12.0 micrograms CG. Remove the tube from the apparatus and examine the gel layer. Record the color and reaction time.

(f) Type IV. Break off the ends of each tube and expose the gel to 10.0 micrograms L. Remove the tube from the apparatus and add 2 drops of sodium hydroxide solution. Examine the gel layer. Record the color and reaction time.

(g) Type V. Break off the ends of each tube and expose the gel to 10.0 micrograms L. Remove the tube from the apparatus and add 2 drops of sodium hydroxide solution. Examine the gel layer. Record the color and reaction time.

(h) Type VI. Break off the ends of each tube and expose the gel to 12.0 micrograms CN. Remove the tube from the apparatus and add 2 drops of sodium hydroxide solution. Examine the gel layer. Record the color and reaction time.

(i) Type VII. Determine the sensitivity as specified in 4.4.4.5(c)(2).

4.4.4.6 Score breakage.

(a) Machine breakage. Determine the score breakage on each end of the tubes using the apparatus shown in figure 1.

(b) Hand breakage. Break both ends of the tubes at the score mark by hand. Observe for compliance with 3.8.

5. PACKAGING

5.1 Unit packing, level C. The detector tube assemblies of the same type and size and of the specified quantity per clip (see 6.2) shall be unit packed to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use.

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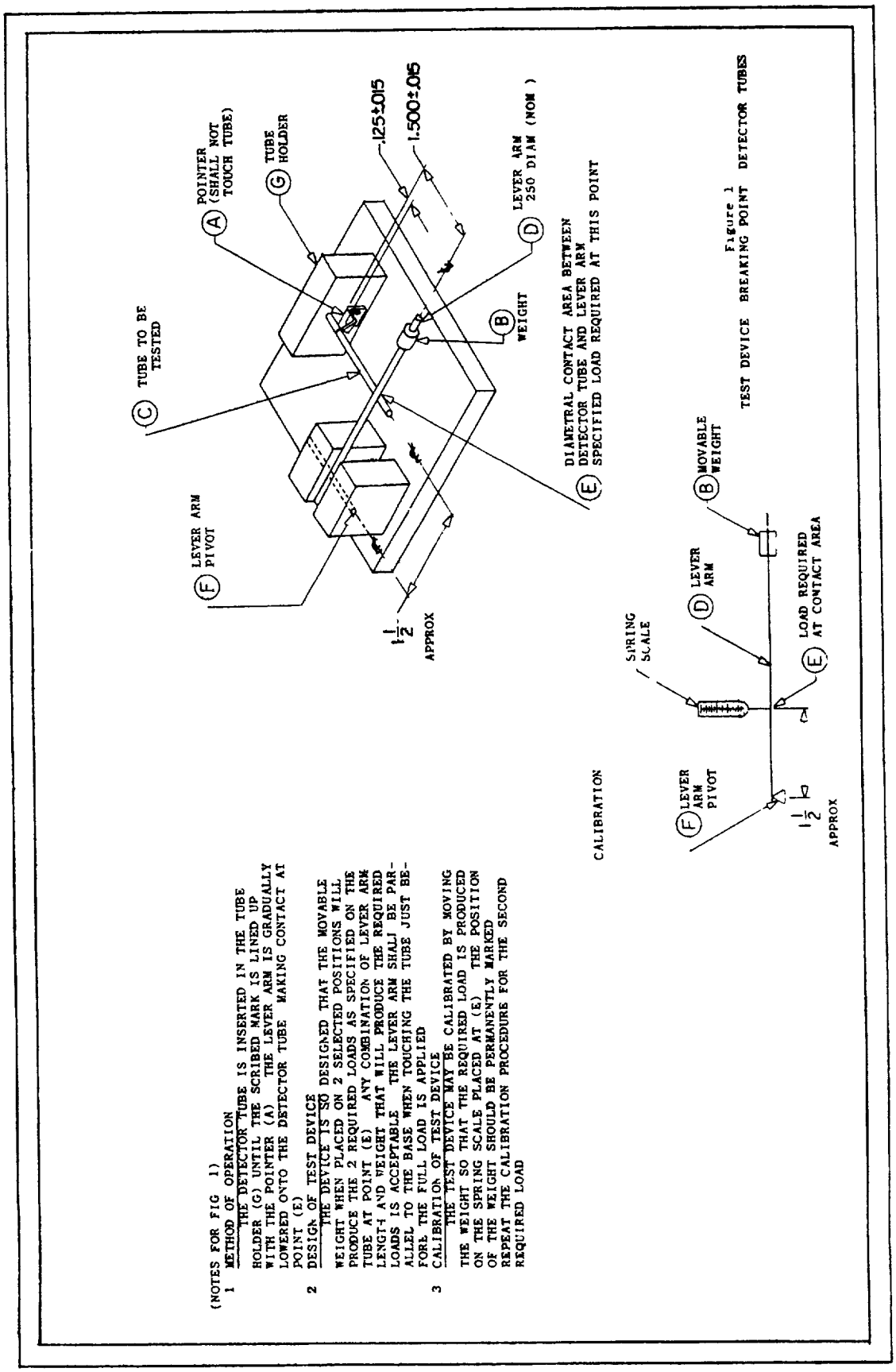


Figure 1

(NOTES FOR FIG 1)

1 METHOD OF OPERATION
THE DETECTOR TUBE IS INSERTED IN THE TUBE HOLDER (G) UNTIL THE SCRIBED MARK IS LINED UP WITH THE POINTER (A) THE LEVER ARM IS GRADUALLY LOWERED ONTO THE DETECTOR TUBE MAKING CONTACT AT POINT (E)

2 DESIGN OF TEST DEVICE
THE DEVICE IS SO DESIGNED THAT THE MOVABLE WEIGHT WHEN PLACED ON 2 SELECTED POSITIONS WILL PRODUCE THE 2 REQUIRED LOADS AS SPECIFIED ON THE TUBE AT POINT (E) ANY COMBINATION OF LEVER ARM LENGTH AND WEIGHT THAT WILL PRODUCE THE REQUIRED LOADS IS ACCEPTABLE THE LEVER ARM SHALL BE PARALLEL TO THE BASE WHEN TOUCHING THE TUBE JUST BEFORE THE FULL LOAD IS APPLIED

3 CALIBRATION OF TEST DEVICE
THE TEST DEVICE MAY BE CALIBRATED BY MOVING THE WEIGHT SO THAT THE REQUIRED LOAD IS PRODUCED ON THE SPRING SCALE PLACED AT (E) THE POSITION OF THE WEIGHT SHOULD BE PERMANENTLY MARKED REPEAT THE CALIBRATION PROCEDURE FOR THE SECOND REQUIRED LOAD

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5.2 Packing level C. Detector tube assemblies, unit packed as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. Unit packs, packs, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When type I and VII detector tubes are procured for storage and issue as a repair part, preservation, unit packing, packing and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. The items covered by this specification are intended for use in the sampling and detection of chemical agents.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Applicable assembly drawing (see 3.2),
(c) Quantity per clip required (see 5.1), and
(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract,
(2) Name and address of test facility and shipping instructions when testing is performed by the Government, and
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Testing precautions. This specification covers the use of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials. This specification requires testing with Chemical Surety Material (CSM) which is potentially hazardous to personnel. Testing with CSM (e.g. L, GS, H, etc.) shall be conducted in a facility which has been approved for this type of operation by the US Army. All general and specialized handling procedures including those in the contract clause "Safety Provisions for Contracts Involving CSM" must be followed when handling, using, or storing CSM. Additionally, this specification requires the use of a known carcinogen.

6.4 Manufacture and protection of gels and unsealed tubes. The description of manufacture for Detector Kit, Chemical Agent, ABC-M18A2 gives suggested methods for preparing types I, II, III, IV, and VII gels and tubes.

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6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

Custodians:

Army - EA
Navy - SH
Air Force - 99

Review activity:

Air Force - 82

User activities:

Navy - MC, OS

Preparing activity:

Army - EA

Project No. 6810-0462

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-51082E	DETECTOR TUBES, CHEMICAL AGENT

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording.

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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